

# A smooth-measure counterexample to the direct generalization of Theorem 3.4

A scoped negative answer to Question 6.1 of arXiv:2510.00645

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## Abstract

Malliaris and Marín Sola ask whether their weighted lower bound and a corresponding upper bound extend to general measures, concavity transforms, and functions  $N$ , explicitly highlighting discrete measures. We show that the direct lower-bound extension is false, even while retaining  $\psi = \log$  and the affine function  $N(t) = t$ . An exact three-atom example already violates the formula after the most natural fractional-atom repair. Replacing the atoms by three narrow translates of one smooth bump preserves the strict violation. Thus there is a counterexample whose measure has a compactly supported  $C^\infty$  density, so neither atom conventions nor a discontinuous distribution function cause the failure. Alternative measure-dependent formulations and the upper-bound part remain open.

## 1 Source question and the direct extension

The source is A. Malliaris and F. Marín Sola, *Quantitative improvements of functional inequalities under concavity properties*, arXiv:2510.00645 [1]. Question 6.1 on PDF page 17 asks whether Theorem 3.4 and the corresponding upper bound hold for general  $\mu, \psi, N$ ; the following paragraph specifically names discrete  $\mu$  as an untreated case.

$e^{-\sqrt{2} \frac{u}{\|x\|_2}}$ , which is not possible in such generality.

**6.2. Towards a general theorem.** Let  $\psi$  be an increasing function defined on  $[0, +\infty)$ ,  $\mu$  a positive Borel measure on  $[0, +\infty)$  and consider the class

$$\mathcal{F}_{h,u,V}^\psi(\mu) := \{f : [0, +\infty) \rightarrow [0, +\infty), \psi\text{-concave, decreasing, } \int_0^h f d\mu = u, \int_0^{+\infty} f d\mu = V\}.$$

As it may be evident from Lemma 2.3 (but also Lemmas 2.2 and 2.1), optimizing functionals of the form

$$\int N f d\mu,$$

over the family  $\mathcal{F}_{h,u,V}^\psi(\mu)$  (for a monotone function  $N$ ) can be reduced to the much smaller families of either  $\psi$ -linear or the combination of constants with  $\psi$ -linear. We are thus naturally led to the following:

**Question 6.1.** Can Theorem 3.4 and the corresponding upper bound be proven for general  $\mu, \psi$  and  $N$ ?

In view of the applications discussed in Sections 4 and 5 many sub-cases of our question seem already interesting. For instance, obtaining a positive answer in the case where  $N(t) = t^2$  and  $\mu$  is the Lebesgue measure would yield a new result, even in the case of  $h \rightarrow 0^+$ . It is also natural to wonder about the optimal upper bound when  $\psi = \log$  and  $\mu$  is the Lebesgue measure, thereby generalizing Theorem 1.1. Furthermore, extending Theorem 3.4 for more general measures remains an open direction. An instance not addressed in this work is when  $\mu$  is a discrete measure, which might be of independent interest; to our knowledge, results of this nature are not available in that setting.

Finally, finding sharp constants of the acceptable ratio  $\frac{u}{V}$  (in the terminology of Section 3) is an important first step toward understanding the behavior of the inequality in the rest regimes of  $\frac{u}{V}$ . Apart from novel inequalities and deeper understanding of the behavior of one-dimensional log-concave functions, such result can bring new insights to the geometric question of finding which slabs –of intermediate volume– are extremal.

For the measures treated in Theorem 3.4, let

$$\Phi(t) = \mu([0, t]), \quad u = \int_0^h f d\mu, \quad V = \int_0^\infty f d\mu.$$

The theorem states, for increasing convex  $N$ , decreasing log-concave  $f$ , and  $u \leq V/2$ , that

$$\int_0^\infty N(t) f(t) d\mu(t) \geq \frac{u}{\Phi(h)} \int_0^{\Phi^{-1}((V/u)\Phi(h))} N(t) d\mu(t). \quad (1)$$

The direct interpretation of the lower-bound part of Question 6.1 is to retain (1) for an arbitrary positive Borel measure. The counterexample below has a continuous distribution function and a unique relevant quantile, so the inverse in (1) is unambiguous.

## 2 Proof intuition

The proposed extremizer effectively fills the first  $(V/u)\Phi(h)$  units of  $\mu$ -mass at the average height  $u/\Phi(h)$ . A general measure can place a large block of mass just to the right of  $h$ . A decreasing exponential  $f$  discounts that block, while the increasing  $N$  magnifies its location. With three carefully weighted blocks, the total-mass constraint is exact but the putative filled-prefix moment is too large.

The atomic calculation has a gap of  $1/4$ . Smoothing each atom by the same narrow even bump preserves the exponential mass ratios exactly. The only errors are first moments inside intervals of radius  $\varepsilon$ , and their total size is less than the atomic gap when  $\varepsilon < 1/18$ .

### 3 Counterexample with a smooth density

**Theorem 1** (Failure for smooth positive Borel measures). *There exist a finite positive Borel measure  $\mu$  on  $[0, \infty)$  with a compactly supported  $C^\infty$  density, a decreasing integrable log-concave function  $f$ , an increasing convex function  $N$ , and  $h > 0$  such that  $u = V/2$  but the inequality (1) is strictly reversed. Consequently the direct lower-bound extension of Theorem 3.4 to general measures is false, already for  $\psi = \log$  and affine  $N$ .*

*Proof.* Fix  $0 < \varepsilon < 1/18$ , and let  $\rho \in C_c^\infty(\mathbb{R})$  be an even nonnegative probability density supported in  $(-\varepsilon, \varepsilon)$ . On  $[0, \infty)$  define

$$d\mu(t) = m(t) dt, \quad m(t) = \rho(t-1) + \rho(t-2) + 6\rho(t-3).$$

Take

$$f(t) = 2^{1-t}, \quad N(t) = t, \quad h = \frac{5}{2}.$$

The supports of the three bumps are disjoint and lie in  $(0, \infty)$ . Clearly  $f$  is decreasing, Lebesgue integrable, and log-concave (indeed,  $\log f$  is affine), while  $N$  is nonnegative, increasing, and convex.

Put

$$A = \int_{-\varepsilon}^{\varepsilon} 2^{-s} \rho(s) ds > 0, \quad B = \int_{-\varepsilon}^{\varepsilon} s, 2^{-s} \rho(s) ds.$$

Since  $|s| < \varepsilon$  on the support,  $|B| \leq \varepsilon A$ . Translation of the same bump and the identity  $f(j+s) = 2^{1-j} 2^{-s}$  give

$$u = \int_0^h f d\mu = A + \frac{A}{2} = \frac{3A}{2}, \quad V = A + \frac{A}{2} + 6\frac{A}{4} = 3A = 2u.$$

Also  $\Phi(h) = 2$ , since precisely the first two unit-mass bumps lie to the left of  $h$ . Hence the mass level in (1) is

$$\frac{V}{u} \Phi(h) = 4.$$

Let  $d = \Phi^{-1}(4)$ . The point  $d$  lies inside the third bump, and if  $q = d - 3$ , then

$$\int_{-\varepsilon}^q \rho(s) ds = \frac{1}{3}.$$

Writing  $C = \int_{-\varepsilon}^q s \rho(s) ds$ , we have  $C \geq -\varepsilon/3$ . Evenness of  $\rho$  makes its full first moment zero, so the prefix  $N$ -moment is

$$\begin{aligned} Q &:= \int_0^d t d\mu(t) = 1 + 2 + 6 \int_{-\varepsilon}^q (3+s) \rho(s) ds \\ &= 9 + 6C \geq 9 - 2\varepsilon. \end{aligned} \tag{1}$$

Thus the proposed right-hand side satisfies

$$\frac{u}{\Phi(h)} Q = \frac{3A}{4} Q \geq \frac{27A}{4} - \frac{3A}{2} \varepsilon. \tag{2}$$

The actual left-hand side can be evaluated on the three translated bumps:

$$\begin{aligned} \int_0^\infty t f(t) d\mu(t) &= (A+B) + \frac{1}{2}(2A+B) + \frac{6}{4}(3A+B) \\ &= \frac{13A}{2} + 3B \leq \frac{13A}{2} + 3\varepsilon A. \end{aligned} \tag{2}$$

Subtracting the upper bound in (2) from the lower bound in (2) gives

$$A\left(\frac{1}{4} - \frac{9}{2}\varepsilon\right) > 0.$$

Therefore the right-hand side of (1) is strictly larger than its left-hand side, proving the claimed failure.  $\square$

## 4 Exact atomic core and robustness

The numerical choices become transparent before smoothing. Let

$$\mu_0 = \delta_1 + \delta_2 + 6\delta_3, \quad f(t) = 2^{1-t}, \quad N(t) = t, \text{quad}2 < h < 3.$$

Then  $\Phi(h) = 2$ ,  $u = 3/2$ ,  $V = 3$ , and

$$\int Nf d\mu_0 = 1 + 2\left(\frac{1}{2}\right) + 6 \cdot 3\left(\frac{1}{4}\right) = \frac{13}{2}.$$

The standard generalized inverse gives  $\Phi^{-1}(4) = 3$  and includes the whole atom at 3, yielding the much larger proposed right side  $63/4$ . More importantly, even if that atom is split fractionally, the first four units of measure have  $N$ -moment

$$1 + 2 + 2 \cdot 3 = 9,$$

so the repaired right side is  $27/4 > 13/2$  by exactly  $1/4$ .

There is also a one-parameter family. For  $0 < r < 1$ , replace  $f$  by  $r^{t-1}$  and the last weight 6 by  $(1+r)/r^2$ . Again  $V = 2u$ , and the fractional-prefix right side minus the actual moment is exactly  $(1-r)/2 > 0$ . The selected value  $r = 1/2$  merely keeps every quantity simple.

## 5 Verification, scope, and novelty guardrail

The script `code/verify_counterexample.py` uses rational arithmetic to check the atomic example, the fractional-atom gap, the entire rational parameter family sampled at  $r = k/100$ , and the rigorous smooth-error margin at  $\varepsilon = 1/100$ . It reports that all checks pass. The smooth proof itself is symbolic and does not depend on computation.

This disproves only the natural same-form lower bound. It neither rules out a regularity- or measure-dependent correction nor settles the upper-bound questions. The smooth density shows that excluding atoms is insufficient.

On 11 August 2026, bounded novelty checks covered the four run indexes and arXiv/web searches by exact title, id, Question 6.1, authors, and “general/discrete measure.” They found the source but no later answer or matching counterexample; this is not a priority claim.

**Human-review recommendation.** Accept as a scoped counterexample to the direct lower-bound half of Question 6.1. Check that retaining (1) is the intended interpretation; all hypotheses and the strict gap are then explicit.

## References

- [1] A. Malliaris and F. Marín Sola, *Quantitative improvements of functional inequalities under concavity properties*, arXiv:2510.00645 (2025), Question 6.1 and Theorem 3.4.